

PHY101: Mechanics

Mid-Semester Examination 2025

Solutions

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Section A

1.

$$F(r) = -\frac{k}{r^2} + \frac{\alpha}{r^3}, \quad (k, \alpha > 0)$$

For very small r , the r^{-3} term dominates because it diverges faster. Since this term is positive (repulsive), the short-range behavior is repulsive.

Answer: (C) — Strongly repulsive at short range.

2. Two smooth beads of masses m and $2m$ connected by a light spring on a frictionless table experience only internal spring forces. By Newton's third law, internal forces sum to zero. Thus the center of mass remains at rest.

Answer: (A) — Center of mass remains stationary.

3. For a free particle moving with constant velocity $\mathbf{v}$,

$$\frac{d\mathbf{L}_O}{dt} = \mathbf{r} \times \mathbf{F} = 0.$$

Hence the angular momentum about any fixed point is constant in magnitude and direction.

Answer: (A) — Constant in both magnitude and direction.

4. The two-body gravitational system is isolated; both bodies orbit their common center of mass with conserved total momentum.

Answer: (B) — Each revolves around the center of mass.

5. In the upward-accelerating lift, a downward pseudoforce ma acts in addition to gravity, giving an effective acceleration $g + a$ downward. Hence, the ball follows a curved path and strikes the opposite wall below the release height.

Answer: (B) — Below the point of release.

Section B

Q6. Central Potential $V(r) = -\frac{k}{r} + \frac{\alpha}{r^2}$

We analyze the motion in this potential for a particle of mass m .

Effective potential:

$$V_{\text{eff}}(r) = \frac{L^2}{2mr^2} - \frac{k}{r} + \frac{\alpha}{r^2} = \frac{A}{r^2} - \frac{k}{r}, \quad A = \frac{L^2}{2m} + \alpha.$$

Condition for circular orbits: A circular orbit at $r = r_c$ satisfies

$$\frac{dV_{\text{eff}}}{dr} = 0 \quad \Rightarrow \quad -\frac{2A}{r_c^3} + \frac{k}{r_c^2} = 0.$$

Hence

$$r_c = \frac{2A}{k}.$$

For $r_c > 0$, both A and k must have the same sign. Assuming $k > 0$, we require $A > 0$.

Stability of the circular orbit: Compute

$$\frac{d^2V_{\text{eff}}}{dr^2} = \frac{6A}{r^4} - \frac{2k}{r^3}.$$

At $r = r_c$,

$$\left. \frac{d^2V_{\text{eff}}}{dr^2} \right|_{r_c} = \frac{6A}{r_c^4} - \frac{2k}{r_c^3}.$$

Using $k = 2A/r_c$ (from above),

$$\left. \frac{d^2V_{\text{eff}}}{dr^2} \right|_{r_c} = \frac{2A}{r_c^4}.$$

Therefore the orbit is stable if $A > 0$.

Critical value of α : Setting $A = 0$,

$$\boxed{\alpha_c = -\frac{L^2}{2m}}.$$

Thus, for $\alpha > \alpha_c$ the orbit is stable; for $\alpha \leq \alpha_c$, it is unstable.

Comment: The additional $+\alpha/r^2$ term modifies the centrifugal barrier; a sufficiently negative α removes stability.

Q7. Motion with Quadratic Drag $F_d = -mk^2v^2$

A body of mass m is projected vertically upward with initial speed v_0 . We find the speed v_1 on returning to the same height.

Upward motion: Equation of motion:

$$m \frac{dv}{dt} = -mg - mk^2 v^2 \quad \Rightarrow \quad v \frac{dv}{dy} = -g - k^2 v^2.$$

Integrate from $v = v_0$ at $y = 0$ to $v = 0$ at $y = h$:

$$h = \int_0^{v_0} \frac{v dv}{g + k^2 v^2} = \frac{1}{2k^2} \ln \left(\frac{g + k^2 v_0^2}{g} \right).$$

Downward motion: Now $v > 0$ downward, forces: $mg - mk^2 v^2$. Equation:

$$v \frac{dv}{dy} = g - k^2 v^2.$$

Integrate from $v = 0$ to $v = v_1$:

$$h = \int_0^{v_1} \frac{v dv}{g - k^2 v^2} = \frac{1}{2k^2} \ln \left(\frac{g}{g - k^2 v_1^2} \right).$$

Equating heights:

$$\frac{1}{2k^2} \ln \left(\frac{g + k^2 v_0^2}{g} \right) = \frac{1}{2k^2} \ln \left(\frac{g}{g - k^2 v_1^2} \right).$$

Simplify:

$$(g + k^2 v_0^2)(g - k^2 v_1^2) = g^2.$$

Expanding and rearranging:

$$g(v_0^2 - v_1^2) = k^2 v_0^2 v_1^2.$$

Solve for v_1^2 :

$$\boxed{v_1^2 = \frac{g v_0^2}{g + k^2 v_0^2}}.$$

Checks:

- For $k \rightarrow 0$: $v_1 = v_0$, as expected (no drag).
- $v_1 < v_0$ for any $k > 0$.

Q8. Motion in Polar Coordinates: $r(t) = r_0(1 + \beta t)$, $\dot{\theta} = \omega_0$

Given $r(t) = r_0(1 + \beta t)$, $\dot{\theta} = \omega_0 = \text{constant}$.

Kinematics in polar form:

$$\mathbf{v} = \dot{r} \hat{r} + r \dot{\theta} \hat{\theta}, \quad \mathbf{a} = (\ddot{r} - r \dot{\theta}^2) \hat{r} + (r \ddot{\theta} + 2\dot{r} \dot{\theta}) \hat{\theta}.$$

Compute derivatives:

$$\dot{r} = r_0 \beta, \quad \ddot{r} = 0, \quad \dot{\theta} = \omega_0, \quad \ddot{\theta} = 0.$$

Hence

$$\begin{aligned} v_r &= r_0 \beta, & v_\theta &= r_0(1 + \beta t) \omega_0, \\ a_r &= -r_0(1 + \beta t) \omega_0^2, & a_\theta &= 2r_0 \beta \omega_0. \end{aligned}$$

When the velocity makes 45° with $\hat{r}$: Condition $v_\theta = v_r$:

$$r_0(1 + \beta t)\omega_0 = r_0\beta \quad \Rightarrow \quad 1 + \beta t = \frac{\beta}{\omega_0}.$$

Therefore

$$t = \frac{\beta/\omega_0 - 1}{\beta} = \frac{\beta - \omega_0}{\beta\omega_0}.$$

This time is positive only if $\beta > \omega_0$.

Acceleration at that instant: Substitute $1 + \beta t = \beta/\omega_0$ into the acceleration components:

$$a_r = -r_0 \left(\frac{\beta}{\omega_0} \right) \omega_0^2 = -r_0\beta\omega_0, \quad a_\theta = 2r_0\beta\omega_0.$$

Magnitude:

$$|\mathbf{a}| = r_0\beta\omega_0\sqrt{(-1)^2 + 2^2} = r_0\beta\omega_0\sqrt{5}.$$

Angle of $\mathbf{a}$ relative to $\hat{r}$:

$$\tan \phi_a = \frac{a_\theta}{a_r} = -2 \quad \Rightarrow \quad \phi_a = \arctan(-2)$$

Conceptual check: Even though ω_0 is constant, $\mathbf{a}$ has both radial and tangential components because r is changing. This is a good illustration of Coriolis-like coupling between $\dot{r}$ and $\dot{\theta}$.
